RAG Chatbot Architecture & Evaluation Report

1. Document Structure and Chunking Logic

The application implements a robust document preprocessing pipeline designed for PDF documents, specifically tailored for policy and legal text.

Document Loading & Cleaning Documents are loaded using `PyMuPDF` (`fitz`), which extracts text page by page. The raw text undergoes a cleaning phase to remove formatting artifacts that could degrade retrieval quality:

- Standalone page numbers and "Page X of Y" patterns are stripped using Regular Expressions.
- Multiple spaces are collapsed, and excessive consecutive newlines are normalized to a maximum of two.
- Trailing whitespaces are trimmed from each line.

Chunking Strategy The cleaned text is split into overlapping chunks using LangChain's `RecursiveCharacterTextSplitter`. This ensures that chunks remain semantically coherent (sentence-aware) and adhere to optimal sizing for the embedding model.

- **Chunk Size:** 1,000 characters (approximately 250 words).
- **Chunk Overlap:** 150 characters (approximately 40 words) to preserve context across chunk boundaries.
- **Separators Used:** `["\n\n", "\n", ". ", "! ", "? ", "; ", " ", ""]` (evaluated in order of preference to keep paragraphs and sentences intact).
- **Metadata:** Each chunk is stored with associated metadata, including a unique `chunk_id`, `source` file name, `word_count`, and `char_count`.

2. Embedding Model and Vector Database

Embedding Model The system uses the `BAAI/bge-small-en-v1.5` model to generate dense vector embeddings.

- **Dimension:** 384
- **Rationale:** The BGE-small model is highly efficient and offers state-of-the-art performance for its size on semantic search tasks. It provides a strong balance between retrieval accuracy and minimal memory footprint, making it ideal for local deployment.

Vector Database The generated embeddings are stored in **ChromaDB**, utilizing a `PersistentClient` for local disk storage.

- **Distance Metric:** Cosine Similarity (`hnsw:space: cosine`).
- **Storage:** Chunks and their embeddings are batch-inserted (in batches of 100) to prevent memory spikes during indexing.

3. Prompt Format and Generation Logic

Generation Model The application leverages the `mistral:7b-instruct` model, served locally via **Ollama**.

- **Context Window:** Restricted to 2,048 tokens to accommodate hardware limitations (e.g., fitting into 4GB VRAM).
- **Temperature:** Set to `0.1` to strongly favor deterministic, factual, and grounded responses, minimizing creative fabrication.

Two-Stage Retrieval To maximize relevance, the pipeline employs a two-stage retrieval process:

1. **Initial Vector Search:** ChromaDB fetches the top 15 candidates (`RETRIEVAL_TOP_K`) using cosine similarity.
2. **Cross-Encoder Reranking:** The candidates are reranked using `cross-encoder/ms-marco-MiniLM-L-6-v2`, and the top 5 chunks (`RERANK_TOP_N`) are selected to form the final context.

Prompt Format The context is injected into a strict, instruction-tuned prompt template. The prompt explicitly instructs the LLM to rely *only* on the provided context and includes a fallback mechanism:

```
You are a knowledgeable and precise assistant specializing in policy an
Your task is to answer the user's question using ONLY the information f

Rules:
1. Base your answer strictly on the provided context.
2. If the context does not contain enough information, respond with: "T
3. Do not fabricate or infer beyond what is explicitly stated.
4. Be clear, concise, and structured in your response.
5. If listing items, use bullet points for readability.

--- CONTEXT START ---
[Section 1]
<chunk 1 text>
...
--- CONTEXT END ---

Question: <user query>
Answer:
```

4. Example Queries and Responses

The evaluation notebook and standalone pipeline test demonstrate the application's capabilities across different scenarios.

Success Case 1: Direct Information Retrieval

- **Query:** "What personal data is collected from users?"
- **Response:** "Based on the provided documents, the personal information collected from users includes their name, email address, and usage data when they register or use the services."
- **Analysis:** High success. The reranker correctly prioritized the exact chunk mentioning data collection, and the model concisely summarized it.

Success Case 2: Multi-Constraint Retrieval

- **Query:** "How long is user data retained?"
- **Response:** "User data is retained for the duration necessary to fulfill the purposes outlined in this policy, unless a longer retention period is required by law."
- **Analysis:** High success. The model extracted the specific retention policy without hallucinating specific timeframes (e.g., 30 days) that were not present in the text.

Success Case 3: Rights and Procedures

- **Query:** "Can I request deletion of my data?"
- **Response:** "Yes, you have the right to request the deletion of your personal data by contacting our privacy team."
- **Analysis:** High success. Accurately captured user rights and the required procedure.

Failure / Edge Case: Out of Domain

- **Query:** "What is something completely unrelated like quantum physics?"
- **Response:** "The provided documents do not contain sufficient information to answer this question."
- **Analysis: Successful Failure.** The prompt's strict instruction effectively prevented hallucination. The pipeline correctly recognized that the retrieved chunks (which likely had very low cosine similarity) did not contain the answer.

5. Limitations, Hallucinations, and Performance Notes

Hallucination Mitigation The combination of a low temperature (`0.1`), explicit prompt constraints, and the strict fallback rule significantly reduces hallucinations. However, if the source document contains contradictory clauses (e.g., in different sections of a legal policy), the model might synthesize an ambiguous answer.

Model and Hardware Limitations

- **Context Truncation:** The 2,048-token context window limits the amount of context that can be fed to the model. If a query requires synthesizing information spread across more than 5 chunks, the model will not have access to the full picture.
- **Complex Reasoning:** While `mistral:7b-instruct` is highly capable, it may struggle with complex multi-hop reasoning or deeply nested legal interpretations compared to larger models like GPT-4.

Latency and Slow Responses Running the entire pipeline locally introduces latency constraints:

- **Time to First Token (TTFT):** The overhead of embedding the user's query, querying ChromaDB, and notably, running the cross-encoder reranker over 15 chunks, can delay the start of the response.
- **Generation Speed:** The token generation speed is bound by the hardware running Ollama. On machines with limited VRAM (e.g., 4GB), generation can be slow, though the streaming UI (`yield`ing tokens as they are generated) significantly improves the perceived user experience.